

The Influence of Noise on 2D Gaussian Profile Parameters Estimation

Anita Gribl, Davor Petrinović

Faculty of Electrical Engineering and Computing, University of Zagreb

Zagreb, Croatia

anita.gribl@fer.hr, davor.petrinovic@fer.hr

Abstract—The precise estimation of the Gaussian profile parameters plays an important role in many scientific fields. In astronomical images, the 2D Gaussian profiles are good approximations of stellar objects, which are point sources spread in the image plane due to various degradations in the imaging process. The accurate estimation of the 2D Gaussian parameters enables stellar object identification and outlier detection. However, various noise types present in astronomical images complicate the estimation process. Image noise can have an additive or multiplicative nature. The primary cause of additive noise is the so-called readout noise, which has a Gaussian distribution and is temperature-dependent. The second noise type is the multiplicative Poisson noise caused by the quantum nature of the light with the standard deviation proportional to the square root of the pixel intensity. Additionally, there are uniformly distributed quantization noise caused by the conversion of the continuous signal to discrete levels and heavy-tailed salt and pepper noise caused by the bad pixels. All these noise types represent the challenge in the precise estimation of Gaussian parameters. Since the iteratively reweighted least squares method (IRWLS) with Huber weights calculated from statistical analysis of estimation errors showed a good performance as a robust estimator in other applications, it is used in this paper for the 2D Gaussian profile parameters estimation from noisy data. The goal of this paper is to analyze the robustness of the IRWLS method to different noise types.

Index Terms—Gaussian profile fitting, noisy data, Huber weights, parameters estimation, iteratively reweighted least squares method, inverse variance weights

I. INTRODUCTION

The precise parameter estimation of the Gaussian function is significant in various research fields such as astrometry [1], spectrography [2], tomography [3], beam characterization [4], etc. In this paper, we are interested in the estimation of the 2D Gaussian profile parameters from actual astronomical images. Stellar objects in astronomical images are point sources spread in the image plane as the consequence of optical aberrations and other degradations that occur during the digital image acquisition process [5]. The spread sources are described by their point spread functions (PSFs). However, the joint action of all these point spread functions, i.e., their convolution in the spatial domain, is well approximated with the 2D Gaussian profile. In the previous paper, we estimated the 2D Gaussian profile parameters of stellar objects from synthesized astronomical images in the case when two or more profiles were overlapped, but in the absence of other noise sources [6]. The iteratively-reweighted least squares (IRWLS) method with

Huber weights was proved successful in solving this problem of overlapping adjacent objects. This paper extends the previous research in [6] by analyzing the efficiency of the IRWLS method in the 2D Gaussian profile estimation from the noisy data. It is important to understand the noise sources, which occur in the actual images during the digital image acquisition process, and their distribution models to determine the optimal weights for the IRWLS method. In this paper, the noise sources are described with the following noise distribution models: additive Gaussian noise, heavy-tailed noise, and Poisson noise. Two types of weights are considered: Huber weights and inverse variance weights. The advantage of Huber weights is that they do not require any prior knowledge about the noise distribution and its variance. Estimation weights are iteratively updated solely on the robust statistics of residual errors. Such weights are optimal for outlier detection and it can be assumed that such weights shall successfully solve the problem of heavy-tailed noise contamination. On the other hand, the inverse variance weights require a priori knowledge of the variance of each sample. However, such weights can be used when the noise contamination follows the Poisson distribution since the sample variance, in that case, is directly related to the noiseless sample value or the model value for that particular sample.

II. RELATED WORK

In [7] and [8], the parameters of the 1D and 2D Gaussian profiles were estimated from the data contaminated with the additive Gaussian noise as the maximum-likelihood solution. Using the Cramer-Rao bound theory, the analytical expressions for parameter variances in the case of Gaussian and Poisson noise contamination were obtained. In [9] and [10], the parameters of the 1D Gaussian profile were also estimated from the data contaminated by the additive Gaussian noise, but in the argument domain. The influence of noise was removed by using the iterative procedure with the known variance of the noise. However, all of those methods require the knowledge of noise variance or noise distribution type. On the other hand, in [6], the IRWLS method with Huber weights was proposed, which does not require any a priori knowledge about the noise distribution, and relies solely on the histogram of estimation errors and its robust statistics. This paper aims to analyze the estimation accuracy of the IRWLS method in the case of data contaminated by different noise types. Moreover, we provide

the optimal weights for the IRWLS method for each noise distribution type.

III. NOISE SOURCES

The digital image acquisition implies the conversion of the light into the digital signal, thereby most commonly using the complementary metal-oxide-semiconductors (CMOS) or charge-coupled devices (CCD) photosensor technologies [11]. The acquisition process consists of three main stages: transform of the incoming light photons into the charge, conversion of the accumulated electrons into the voltage output, and the A/D conversion of the voltage output to the quantized digital pixel value. Each stage involves different types of noise sources that can be divided into two main categories: random noise (time-variant noise) and spatial noise (time-invariant noise, but also time-variant noise) [12].

A. Random noise

The first acquisition stage is determined by the property of the photosensor, known as the quantum efficiency, to produce the charge in the form of electrons from the incoming photons [13]. The random noise sources in the first acquisition stage are the photon shot noise and the dark current shot noise. The discrete nature of the light causes the photon shot noise. Namely, the photon detection on the sensor surface during the exposure period is a random process that follows the Poisson distribution [13]. The next step is the electron generation from the detected photons. The electrons can be generated even in the absence of light, thereby producing the dark current. Dark current shot noise is the consequence of the sensor structural imperfections that cause the random electron generation, which can be also modeled with the Poisson distribution. It increases with the temperature and exposure time, but it is signal-independent. The second acquisition stage is the readout stage where the generated electrons are read for each pixel as the difference of potentials (initial and final after the exposure). This stage involves reset noise and readout noise. The reset noise is caused by the uncertainty of the reference voltage for each pixel. The reference voltage is the voltage in the absence of light. It is set to some positive offset value to ensure that the readout pixel values are always positive. Both reset noise and readout noise are thermally generated and modeled by the Gaussian distribution [14].

B. Spatial noise

The spatially-variant noise involves fixed pattern noise sources, and different CCD and CMOS specific noise sources such as row or column fixed pattern noise, bad pixels, etc [12]. In the CCD sensors, the accumulated charge is transferred across the sensor to the output amplifier where it is converted to the output voltage. On the other hand, in the CMOS sensors, each pixel column has its amplifier for the charge to voltage conversion, thus increasing the spatial fixed pattern noise (FPN). FPN sources include the photo-response non-uniformity (PRNU), which appears in the first acquisition stage, causing the different number of generated electrons

between pixels under the uniform illumination. There is also dark current non-uniformity (DCNU), which causes the variant dark current generation in different sensor pixels [11]. The DCNU can be modeled by the heavy-tailed distribution to account for the outliers or hot pixels [13].

The third phase of A/D conversion involves the simple quantization noise of uniform distribution and due to high resolution, it is typically negligible to other noise sources.

IV. IRWLS METHOD FOR 2D GAUSSIAN PROFILE ESTIMATION

IRWLS method iteratively minimizes the weighted sum of residual errors. The objective function of the IRWLS method is written as

$$D(\beta) = \sum_{i \in ROI} w_i e_i^2, \quad (1)$$

The weights can be calculated in different ways. They can be calculated as a function of residual errors using the so-called M -estimators such as Huber or bi-square estimators [15]. This class of estimators showed good performance if the errors follow the heavy-tailed distribution. They successfully detect the outliers and deweight them. The outliers are detected using the robust statistics provided that at least half of input samples are regular since the 50% of valid samples is the breakdown point of any robust estimator. The robust measures of error dispersion SD can be estimated from the histogram of errors using the median of absolute deviations from the median of the data as $\hat{SD} \approx MAD \cdot 1.4826$ or based on the inter-quartile range $\hat{SD} \approx IQR \cdot 1.349$. The estimated error dispersion is used to determine the threshold δ , which separates the outliers from regular samples. The outliers are the samples with the absolute residual error larger than the determined threshold δ , while other samples are considered valid. Such estimators are especially interesting since they do not require any prior knowledge about the noise distribution or variances of each sample. They use only the robust statistic of errors for the calculation of estimation weights.

A. Huber weights

The first weights considered in this paper are Huber weights. The valid samples have unit weights, while the weights of outliers decrease with increasing absolute residual error. Huber weights are calculated as

$$w_i(e_i) = w_H(e_i) = \begin{cases} 1, & \text{if } |e_i| \leq \delta \\ \delta/|e_i|, & \text{if } |e_i| > \delta, \end{cases} \quad (2)$$

where e_i denotes the residual error of the i th input sample. The threshold δ is calculated as the product of the robust measure of residual errors dispersion $\hat{SD}$ and the coefficient k as

$$\delta = \hat{SD} \cdot k \quad (3)$$

A smaller value of coefficient k yields the solution closer to the L_1 norm and makes the estimator more resilient to outliers. However, such a small value of coefficient k decreases the asymptotical efficiency of the estimator, thus meaning that a

larger number of input samples is required to achieve the same modeling gain for the case without outliers. A larger value of coefficient k yields a solution closer to the L_2 norm, of higher asymptotic efficiency but lower resilience to outliers. In [15], the threshold value δ of $\delta = 1.345\hat{S}D$ is suggested to achieve the desired robustness with sufficiently high asymptotic efficiency (95%) in the case of normally distributed errors. In addition to efficiency, the value of the coefficient k in the threshold δ also determines the convergence rate of the IRWLS method, as we analyzed in [16].

B. Inverse variance weighting

The second weights considered in this paper are the inverse variance weights, which can be calculated as

$$w_i(\sigma_i^2) = w_I(\sigma_i^2) = \frac{1}{\sigma_i^2 \sum_{i=1}^N (\frac{1}{\sigma_i^2})} \quad (4)$$

where the σ_i^2 is the variance of the i th pixel value, and N is the total number of samples within the estimation region of interest (ROI). For the case of Poisson distributed noise, the variance of each sample is equal to the noiseless sample value or sample value approximated by the model as

$$\sigma_i^2 = f(\mathbf{v}_i, \hat{\beta}) \quad (5)$$

In the case of pure Poisson distributed noise, estimation using the inverse variance weighting with weights obtained by the model values will converge to the optimal solution provided that a model obtained in the particular iteration is close enough to the actual solution. Therefore, sufficiently close initialization of the iterative procedure is required to achieve convergence. If somehow the variances of each sample are known in advance, the iterative method becomes a simple weighted least squares method (WLS) with the one-step solution.

C. 2D Gaussian profile

The normalized 2D Gaussian profile with linear scale A_l can be expressed as

$$f(\mathbf{v}_i) = \frac{A_l}{2\pi\sqrt{\det(\mathbf{C})}} \exp \left[-\frac{1}{2} \left((\mathbf{v}_i - \boldsymbol{\mu})^T \mathbf{C}^{-1} (\mathbf{v}_i - \boldsymbol{\mu}) \right) \right], \quad (6)$$

where A_l denotes the amplitude of the profile, $\mathbf{v}_i = [x_i \ y_i]^T$ denotes the position of the i th pixel within the ROI, $\boldsymbol{\mu} = [\mu_x \ \mu_y]^T$ denotes the peak position of the profile and $\mathbf{C}$ denotes the covariance matrix. The readout value of the i th pixel is denoted with z_i . The estimation ROI involves the input samples' positions and their corresponding values that are passed to the optimization process. In our experiments, the ROI was chosen as a convex input region of the samples within the Mahalanobis distance of radius k . The Mahalanobis distance is defined as a square root of the quadratic form in the argument of the Gaussian function. If the widths of major and minor semiaxes are the same $\lambda_1 = \lambda_2 = \lambda$, the input region of the samples within the Mahalanobis distance of $k\lambda$ is a convex circular region. If the major and minor semiaxes have different lengths, the radius of Mahalanobis distance k determines the

elliptical input region of width $k\lambda_1$ in the direction of the major and $k\lambda_2$ in the direction of the minor semiaxis.

The covariance matrix is obtained from an orthonormal rotation matrix $\mathbf{R}$ and a diagonal matrix of eigenvalues $\mathbf{S}$ as follows:

$$\mathbf{R} = \begin{bmatrix} \cos(\frac{\theta}{2}) & -\sin(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{bmatrix}, \mathbf{S} = \begin{bmatrix} \lambda_1^2 & 0 \\ 0 & \lambda_2^2 \end{bmatrix}, \mathbf{C} = \mathbf{R}\mathbf{S}\mathbf{R}',$$

where $\frac{\theta}{2} = \theta$ is the rotation angle.

D. Objective function

The vector of parameters for the estimation process is

$$\boldsymbol{\beta} = [A_l \ \lambda_1 \ \lambda_2 \ \mu_x \ \mu_y \ \rho]. \quad (7)$$

If z_i is the value of the i th pixel and its position is $\mathbf{v}_i = [x_i \ y_i]$, the error of the model $f(\mathbf{v}_i, \boldsymbol{\beta})$ for the i th pixel is

$$e_i = z_i - f(\mathbf{v}_i, \hat{\beta}). \quad (8)$$

If the weights for all samples involved in the optimization process are equal, the IRWLS method becomes the least squares (LS) method. If the weights are calculated as Huber weights, the IRWLS method becomes robust and resilient to outliers as it was shown in [6]. The application of the IRWLS method with Huber weights [17] represents the soft thresholding since the irregular samples are not discarded but are nevertheless taken into account with reduced weights. The Huber weights are calculated from the residual errors without any knowledge about the noise distribution or samples' variances. Contrary, the LS method is highly affected by outliers since it treats each sample with the same significance. However, the LS solution represents the Cramer-Rao bound in the case of additive Gaussian noise since the mean squared error (MSE) solution is the maximum-likelihood (MLE) solution for this type of contamination [17].

E. IRWLS algorithm

The IRWLS method iteratively minimizes the objective function in (1) to find the optimal vector of estimated parameters $\hat{\beta}$ as

$$\hat{\beta} = \arg \min_{\beta} D(\beta). \quad (9)$$

The pseudocode of the IRWLS method with Huber weights and a detailed algorithm description are given in [6]. The IRWLS algorithm consists of two loops. The outer loop iteratively modifies weights according to the residual errors from the previous iteration, while the inner loop performs the nonlinear optimization with the fixed weights calculated in the outer loop. The algorithm terminates when the absolute threshold change in two consecutive iterations is less than the specified optimization termination tolerance or if the number of outer loop iterations achieved the given bound. Initial weights are set to unit values and the solution obtained in the first iteration is the simple LS solution. The certainty of finding a global optimum of (9) depends on the closeness of the initial model to the optimal solution. The gradients of the objective function in (1) can be calculated numerically during

the optimization or can be evaluated from analytical expressions and passed to the optimization process. The analytical derivatives accelerate the optimization process and they can be also found in [6].

V. EXPERIMENTS AND RESULTS

A. Estimation of the 2D Gaussian profile

The first experiment involves the estimation of one synthesized 2D Gaussian profile from the noisy data. Two different simulations were made. The heavy-tailed noise was added to the noiseless model in the first simulation, while the Poisson distributed noise was applied to the noiseless model in the second simulation. For both simulations, the chosen signal-to-noise ratio (SNR) was set to 30 dB. The ROI was chosen as an elliptical region within the radius of Mahalanobis distance $k = 3$. The specified model parameters were passed to the optimization process as the initial model, to circumvent the problem of appropriate initialization that will not be considered here. The synthesized profile was positioned at the center of the synthetic image, but with the random fractional shift within the half-pixel region to mimic the discrete sampling on the uniform grid. The major semiaxis was two times larger than the minor semiaxis ($\lambda_1 = 2, \lambda_2 = 1$), the rotation angle was $\theta = \rho/2 = \pi/8$, while the linear amplitude was set to $A_l = 10^5$. The maximum number of outer loop iterations t_{max} was 50, and the optimization termination tolerance was set to $tol = 10^{-10}$. The appropriately scaled noise was added to the samples within the ROI such that the total energy of the signal in the ratio with the total energy of scaled noise within the ROI satisfies the required signal-to-noise ratio in decibels. If $\bar{z}_i$ denotes the noiseless signal value of the i th pixel within the ROI, n_i denotes the noise sample added to the i th pixel value and α denotes the noise scale factor, the total signal can be written as:

$$z_i = \bar{z}_i + \alpha n_i \quad (10)$$

with the SNR given by the following expression:

$$SNR = 10 \log_{10} \frac{\sum_{i \in ROI} \bar{z}_i^2}{\sum_{i \in ROI} \alpha^2 \cdot n_i^2} \text{ [dB]} \quad (11)$$

To compare the accuracy of the methods for the 2D Gaussian profile estimation, the total modeling error was used, which, as a simple joint scalar measure, shows the accuracy of all model parameters. Total modeling error can be calculated over the evaluation region as

$$r = 10 \log_{10} \frac{\sum_{j \in eval} (f(\mathbf{v}_j, \hat{\beta}) - f(\mathbf{v}_j, \beta))^2}{\sum_{j \in eval} f(\mathbf{v}_j, \beta)^2} \text{ [dB]} \quad (12)$$

where $f(\mathbf{v}_j, \hat{\beta})$ represents the estimated model calculated at pixel position $\mathbf{v}_j$, while $f(\mathbf{v}_j, \beta)$ is the specified model, and $eval$ is the arbitrary evaluation grid. In our experiments, the evaluation region was the same as the estimation region but of higher spatial density to better approximate the integral of the continuous error function over the evaluation region. The 2D Gaussian profile parameters were estimated using the LS method, IRWLS method with Huber weights according to (2),

and IRWLS method with normalized inverse variance weights according to (4). The efficiency of these three methods in the profile estimation from noisy data was compared.

1) *Heavy-tailed noise contamination:* The vector of actual profile parameters for the first experiment was chosen as $\beta = [A_l, \lambda_1, \lambda_2, \mu_x, \mu_y, \rho] = [10^5, 2, 1, -0.3082, -0.2817, 0.7854]$. The process of heavy-tailed noise generation and parameters estimation using the IRWLS method with Huber weights and the WLS method with inverse variance weights are shown in Fig. 1. To generate the noise from heavy-tailed distribution, the MATLAB function *random* was used. The random numbers were generated from the generalized Pareto distribution with the following parameters: the shape parameter $k = 3$, the scale parameter $\sigma = 1$, and the threshold parameter $\theta = 0$. The generated numbers were used to represent the long-term distribution statistics, i.e., the expected variances of noise samples for individual pixels. To generate a random sample that satisfies such spatial distribution of variances, the random samples were drawn from the normal distribution for each pixel within the ROI and their absolute values were multiplied with the square root of the spatially variant long-term expectations of noise samples' variances. The obtained noise samples were finally scaled with an appropriate factor α to achieve the required SNR in dB.

Huber weights are smaller for outliers and obtain unit values for regular samples. The inverse variance weights calculated from the specified spatially-dependent variances shall yield the more optimal solution, but the long-term noise statistics must be known in advance. However, we showed that the Huber weights are successful in solving the problem of heavy-tailed noise, but without any prior knowledge with fairly high estimation accuracy. For comparison, the IRWLS method with Huber weights achieved the total modeling error of -64.2273 dB with the estimated vector of parameters $\hat{\beta}_H = [1.0008 \cdot 10^5, 2.0005, 1.0006, 0.7855, -0.3083, -0.2816, 0.7855]$, the WLS method with the specified inverse variance weights achieved the total modeling error of -82.8051 dB with the estimated vector of parameters $\hat{\beta}_I = [1.0001 \cdot 10^5, 2.0001, 1.0000, -0.3082, -0.2817, 0.7854]$, while the LS method achieved the error of -36.4718 dB with the estimated vector of parameters $\hat{\beta}_{LS} = [1.0136 \cdot 10^5, 1.9950, 1.0175, -0.3199, -0.2864, 0.7823]$. The example of the final absolute residual errors of the estimated models for the IRWLS, WLS, and LS methods are given in Fig. 1.

2) *Poisson noise contamination:* In the second experiment, the Poisson noise was applied to the 2D Gaussian profile instead of heavy-tailed noise. The new vector of specified parameters in this simulation was chosen as $\beta = [A_l, \lambda_1, \lambda_2, \mu_x, \mu_y, \rho] = [10^5, 2, 1, -0.3639, -0.3530, 0.7854]$, where only the center position was changed. The variance of the applied Poisson noise for each sample is equal to the noiseless signal sample value ($\sigma_i^2 = \bar{z}_i$). The bipolar symmetric noise was generated as normally distributed noise of zero mean and

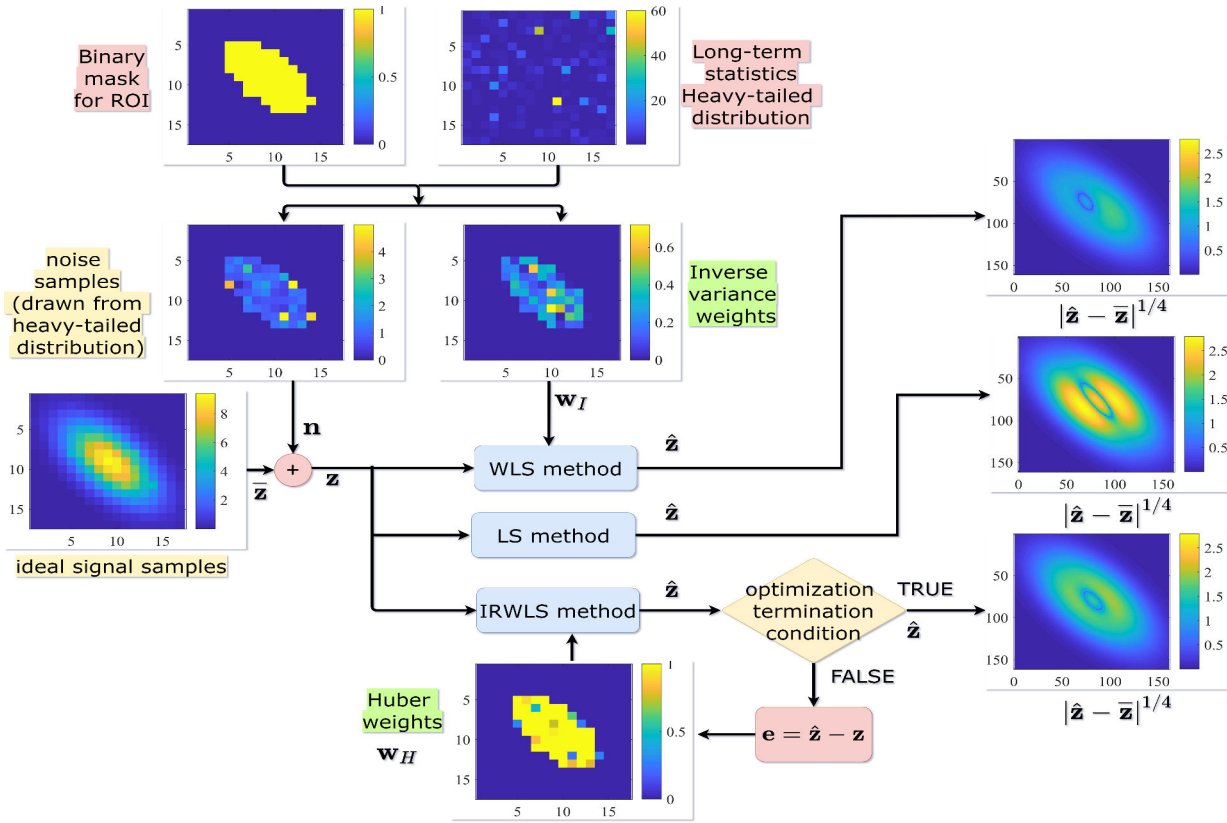
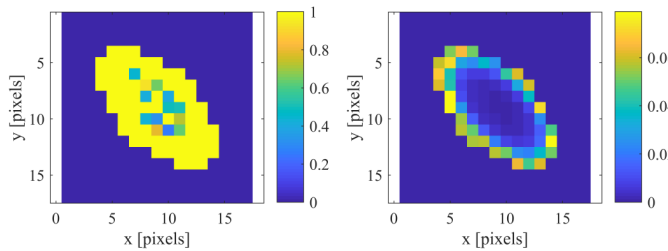


Fig. 1: The IRWLS method with Huber weights and the WLS method with inverse variance weights in the case of heavy-tailed noise contamination

unit variance but multiplied with the square root of the ideal samples' intensities. The noise was also scaled with the factor α to obtain the required SNR. The IRWLS method



(a) Huber weights within the ROI (b) Normalized inverse variance weights within the ROI

Fig. 2: Estimation weights in the last iteration of the IRWLS method when the profile was contaminated with **Poisson noise**. The pixel size was 1 and the sampling grid was uniform. The maximum number of outer loop iterations was 50.

with Huber weights achieved the total modeling error of -36.9081 dB with the estimated vector of parameters $\hat{\beta}_H = [9.9343 \cdot 10^4, 1.9929, 0.9889, -0.3695, -0.3721, 0.7759]$. Unlike the heavy-tailed noise, which is pixel-dependent, Poisson noise is profile-dependent so the IRWLS method had to be applied in which the inverse variance weights were calculated from the current estimate of the model. For

such estimation, the total modeling error of -42.7928 dB was obtained with the estimated vector of parameters $\hat{\beta}_I = [9.9719 \cdot 10^4, 2.0083, 0.9985, -0.3583, -0.3573, 0.7837]$. Finally, the LS method achieved the error of -36.8037 dB with the estimated vector of parameters $\hat{\beta}_{LS} = [9.9374 \cdot 10^4, 1.9961, 0.9931, -0.3678, -0.3734, 0.7739]$. The Huber and inverse variance estimation weights in the last iteration of the IRWLS method are shown in Fig. 2. The inverse variance weights, calculated from the model in each outer loop iteration, reduce the influence of larger samples with larger variances and emphasize the influence of samples at the ROI edge where the variances are much smaller. The variations of weights at the ROI edge are the consequences of coarse discrete sampling and fractional shift of the profile's peak position. The Huber weights are mostly equal to one (regular samples), and only the smaller percentage of pixels around the center of the profile with large estimation error has reduced weights. That is consistent with the characteristics of Poisson contamination but less efficient than the inverse variance weighting. However, the total modeling errors of all three methods differ in only a few decibels, while the inverse variance weights yielding the best result.

B. Experiment 2 - Monte Carlo simulation

By extending the described simple examples and to better analyze the efficiency of the estimation methods, the Monte

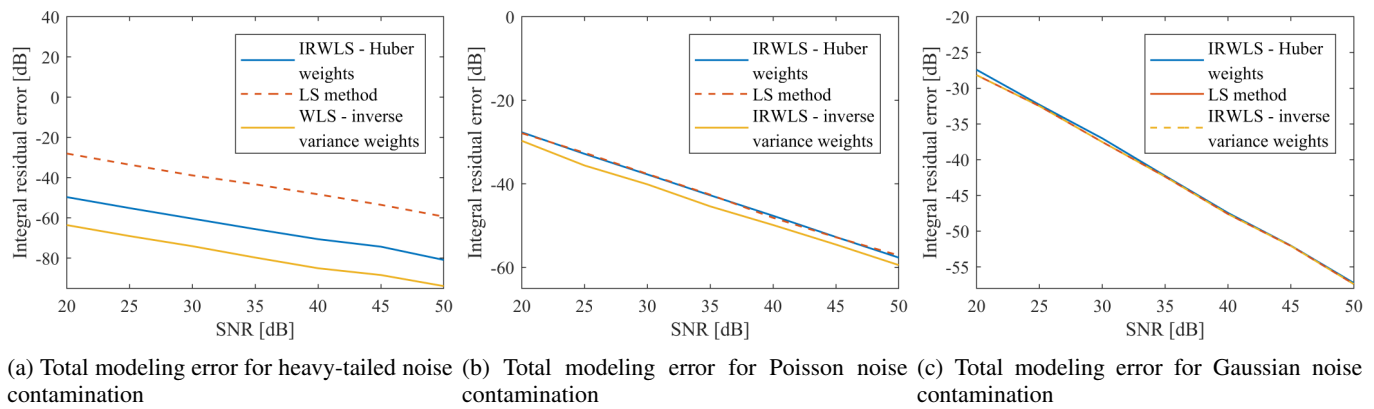


Fig. 3: Comparison of IRWLS method with Huber weights, IRWLS method with inverse variance weights, and LS method for different signal-to-noise ratios (SNR) in dB and different noise distributions.

Carlo simulation was performed. The profile was fixed, but 100 trials were performed with randomized fractional profile shift for all three noise and estimator types and different signal-to-noise ratios in the range from 20 to 50 dB with the step size of 5 dB. The profiles were fractionally shifted to mimic the random discrete sampling on the uniform grid. The results are shown in Fig 3, showing the median of modeling error for 100 trials. In the case of heavy-tailed noise, the IRWLS method with Huber weights yields up to 25 dB smaller total modeling error than the LS method, while the WLS method with inverse weights yields even better, close to the optimal solution. In the case of Poisson noise, the weights were model-derived and iteratively adjusted to the input samples in the IRWLS method. The IRWLS method with inverse variance weights derived from the model yields the smallest modeling error (up to 3 dB better than the LS solution) since such weighting is an MLE solution for the Poisson distributed contamination. For the case of pure Gaussian estimation, the LS method is an optimal solution in MSE and MLE sense, and the inverse variance weighting solution is the same as the LS solution since all pixels variances are chosen to be equal.

VI. CONCLUSION

The main benefit of the IRWLS method with Huber weights in the estimation from the noisy data is that a priori knowledge about the noise distribution is not required. The method can achieve a large modeling gain of up to 25 dB in the case of heavy-tailed noise contamination. Furthermore, the appropriate choice of the threshold δ yields the high asymptotic efficiency of the IRWLS method in the case of additive Gaussian noise contamination. For the case of Poisson distributed noise, the IRWLS method with inverse variance weights has the largest modeling gain of up to 3 dB compared to the IRWLS method with Huber weights and the LS method since the inverse variance weighting is optimal in the MLE sense for this type of contamination. For the case of additive Gaussian noise, the LS solution is optimal as expected.

REFERENCES

- [1] L. Auer and W. Van Altena, "Digital image centering. ii," *The Astrophysical Journal*, vol. 83, pp. 531–537, 1978.
- [2] J. Ireland, "Precision limits to emission-line profile measuring experiments," *The Astrophysical Journal*, vol. 620, no. 2, p. 1132, 2005.
- [3] Y.-C. Chen, L. R. Furenliid, D. W. Wilson, and H. H. Barrett, "Calibration of scintillation cameras and pinhole spect imaging systems," in *Small-Animal SPECT Imaging*. Springer, 2005, pp. 195–201.
- [4] L. Kazovsky, "Beam position estimation by means of detector arrays," *Optical and Quantum Electronics*, vol. 13, no. 3, pp. 201–208, 1981.
- [5] A. Gribl and D. Petrinović, "Synthetic astronomical image sequence generation," in *2020 43rd International Convention on Information, Communication and Electronic Technology (MIPRO)*. IEEE, pp. 1103–1108.
- [6] A. Gribl and D. Petrinović, "A robust method for gaussian profile estimation in the case of overlapping objects," *IEEE Access*, pp. 1–1, 2021.
- [7] N. Hagen, M. Kupinski, and E. L. Dereniak, "Gaussian profile estimation in one dimension," *Applied optics*, vol. 46, no. 22, pp. 5374–5383, 2007.
- [8] N. Hagen and E. L. Dereniak, "Gaussian profile estimation in two dimensions," *Applied optics*, vol. 47, no. 36, pp. 6842–6851, 2008.
- [9] H. Guo, "A simple algorithm for fitting a gaussian function [dsp tips and tricks]," *IEEE Signal Processing Magazine*, vol. 28, no. 5, pp. 134–137, 2011.
- [10] I. Al-Nahhal, O. A. Dobre, E. Basar, C. Moloney, and S. Ikki, "A fast, accurate, and separable method for fitting a gaussian function [tips & tricks]," *IEEE Signal Processing Magazine*, vol. 36, no. 6, pp. 157–163, 2019.
- [11] C. Aguerrebere, J. Delon, Y. Gousseau, and P. Musé, "Study of the digital camera acquisition process and statistical modeling of the sensor raw data," 2013.
- [12] A. J. Theuwissen, "Better pictures through physics," *IEEE Solid-State Circuits Magazine*, vol. 2, no. 2, pp. 22–28, 2010.
- [13] M. Konnik and J. Welsh, "High-level numerical simulations of noise in ccd and cmos photosensors: review and tutorial," *arXiv preprint arXiv:1412.4031*, 2014.
- [14] R. Costantini and S. Susstrunk, "Virtual sensor design," in *Sensors and Camera Systems for Scientific, Industrial, and Digital Photography Applications V*, vol. 5301. International Society for Optics and Photonics, 2004, pp. 408–419.
- [15] J. Fox and S. Weisberg, "Robust regression," *An R and S-Plus companion to applied regression*, vol. 91, 2002.
- [16] A. Gribl and D. Petrinović, "The influence of the huber estimator tuning constant on the performance of the iteratively reweighted least squares method," *Abstract Book, Fifth International Workshop on Data Science, Centre of Research Excellence for Data Science and Cooperative Systems Research Unit for Data Science*, pp. 12–14, 2020.
- [17] P. J. Huber, "Robust estimation of a location parameter: Annals mathematics statistics, 35," 1964.